Contents

Read 2.5, 2.6, 2.9

Artin 2.5.1

Problem 1: Prove that the nonempty fibres of a map form a partition of the domain.

Artin 2.5.6

Problem 2: (a) Prove that the relation x conjugate to y in a group G is an equivalence relation on G.

(b) Describe the elements a whose conjugacy class (= equivalence class) consists of the element a alone.

Artin 2.6.2

Problem 3: Prove directly that distinct cosets do not overlap.

Artin 2.6.4

Problem 4: Give an example showing that left cosets and right cosets of $\mathrm{GL}_2(\mathbb{R})$ in $\mathrm{GL}_2(\mathbb{C})$ are not always equal.

Artin 2.6.5

Problem 5: Let H, K be subgroups of a group G of orders $3, 5$ respectively. Prove that $H \cap K = \{1\}$.